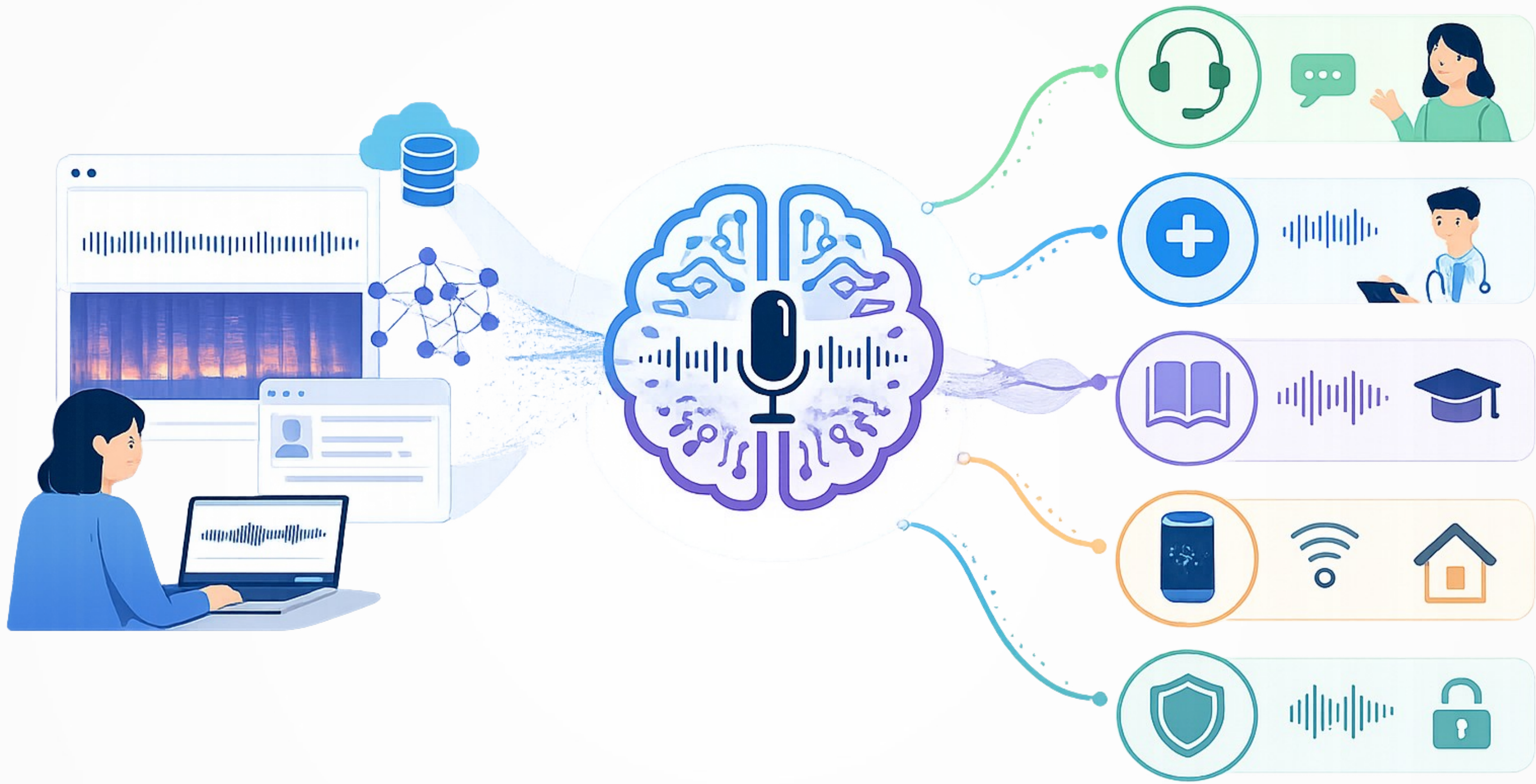
# Voice and Mutimodal AI in Healthcare




Bagus Tris Atmaja

Nara Institute of Science and Technology (NAIST)

Email: bagus.tris@naist.ac.jp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# Overview

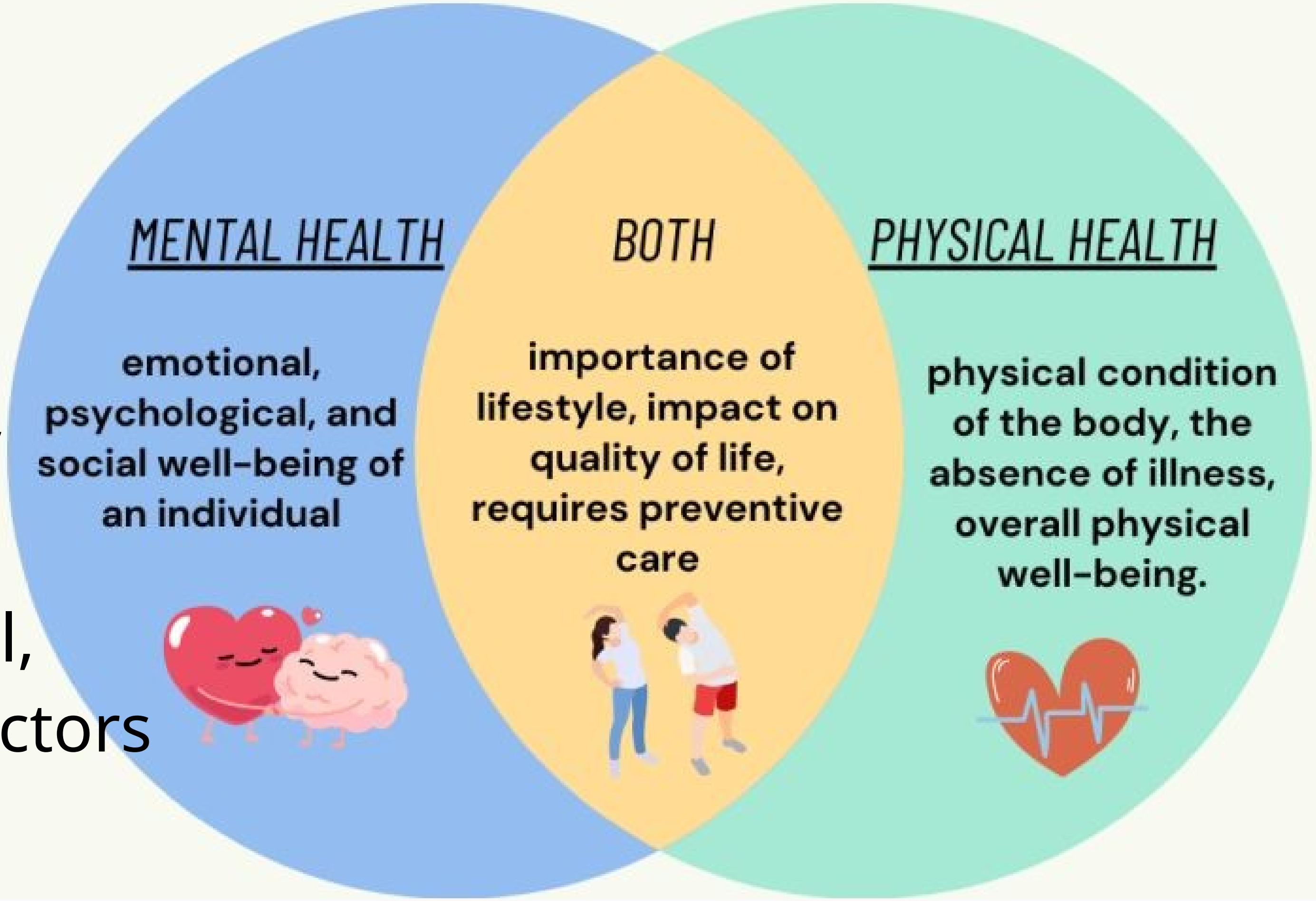
- AI, Deep Learning, Machine Learning
- PART I: VOICE AI
- PART II: MULTIMODAL AI
- Hands-On: **CoughKit, Nkululeko, MultiBench**

# Health, healthcare and well-being

Healthcare → improvement or maintenance of health via the prevention, diagnosis, treatment, rehabilitation, and palliative care (easing suffering)

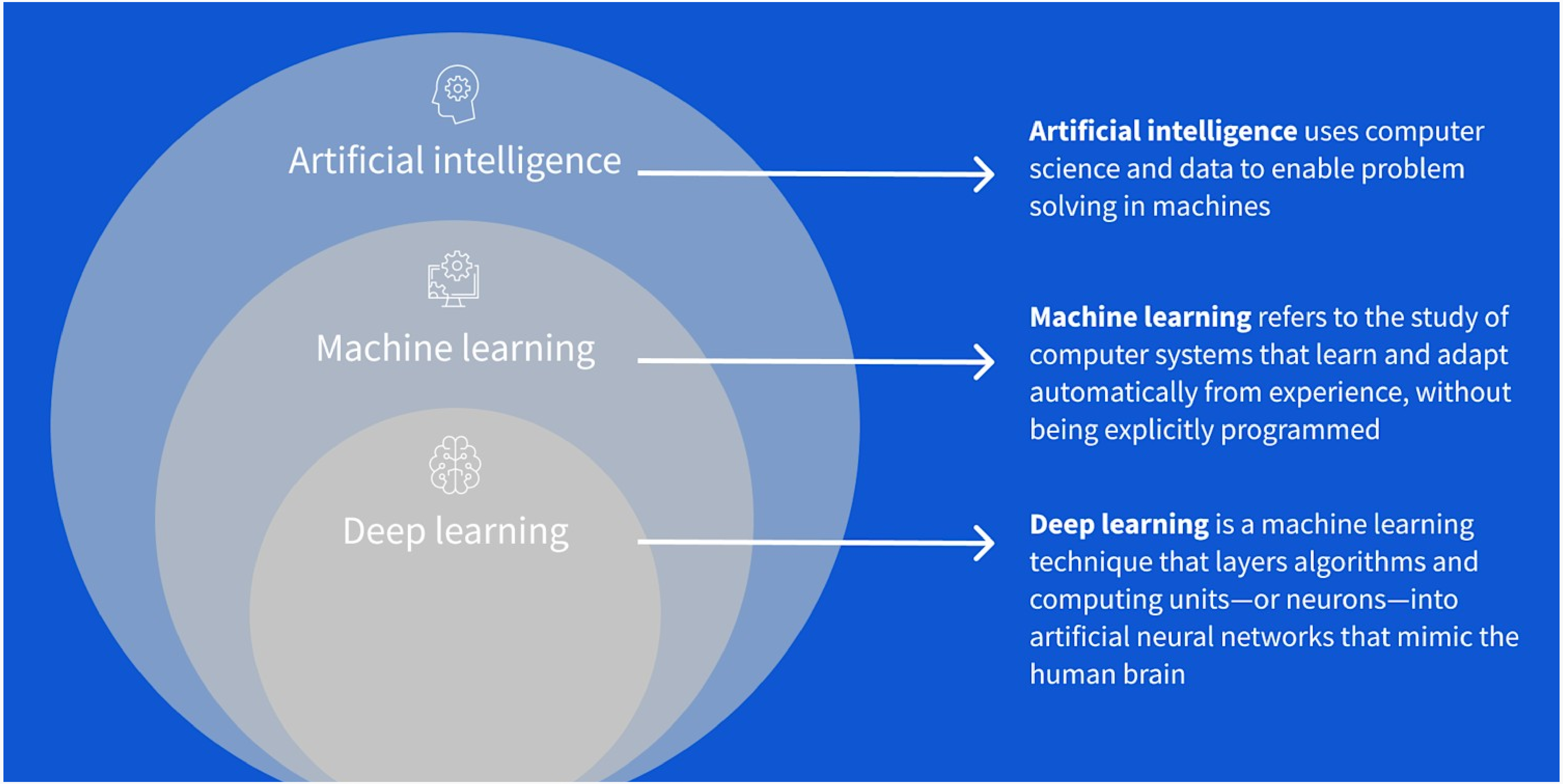
Health
- **Physical**
- Mental

**Well-being**: a complex state of feeling good about life, encompassing physical, mental, emotional, and social health factors



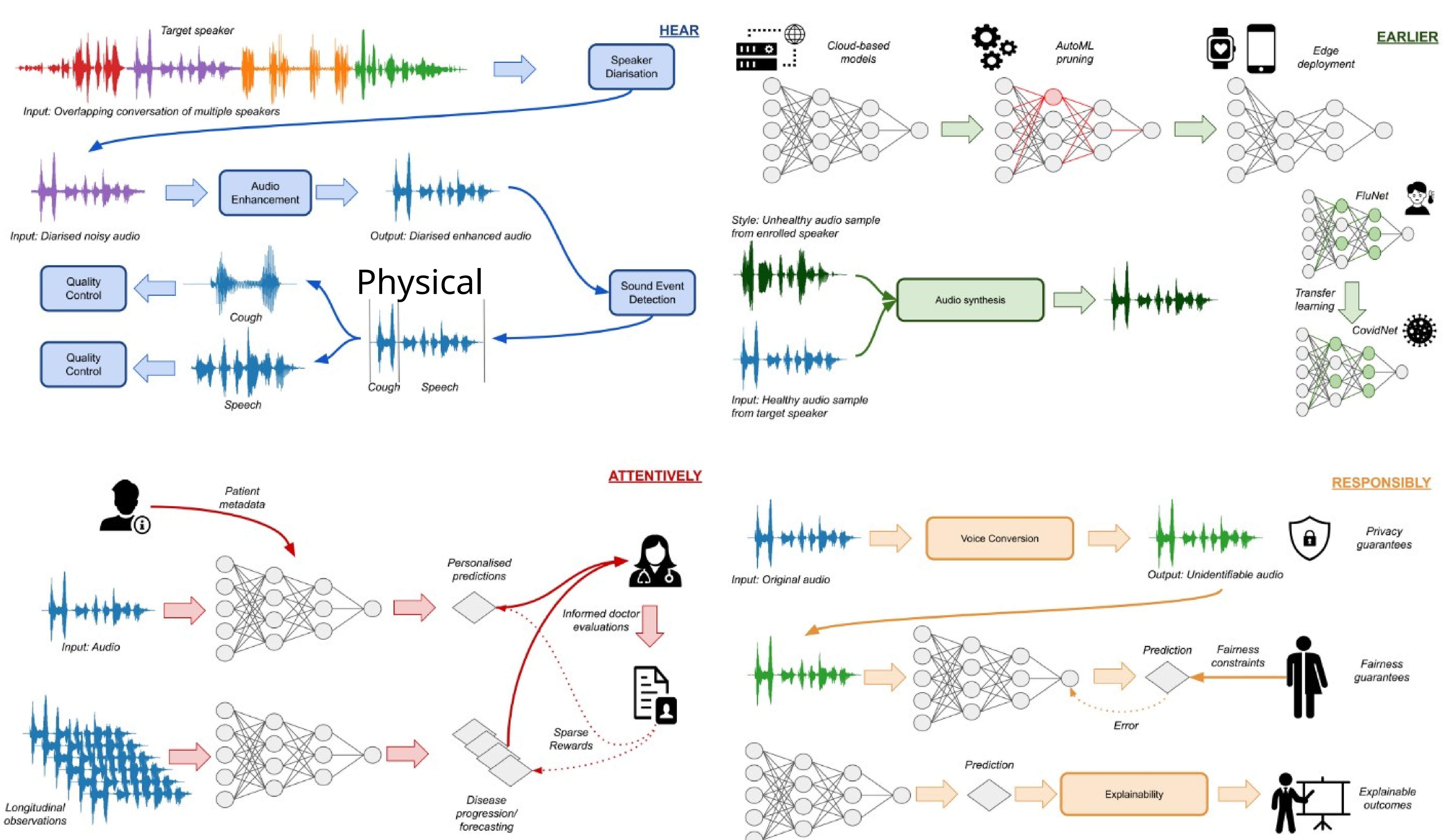

https://www.linkedin.com/posts/the-mind-and-company_venndiagramwednesdays-mentalhealth-physicalhealth-activity-7226947436605292545-fSou

# AI, ML, DL

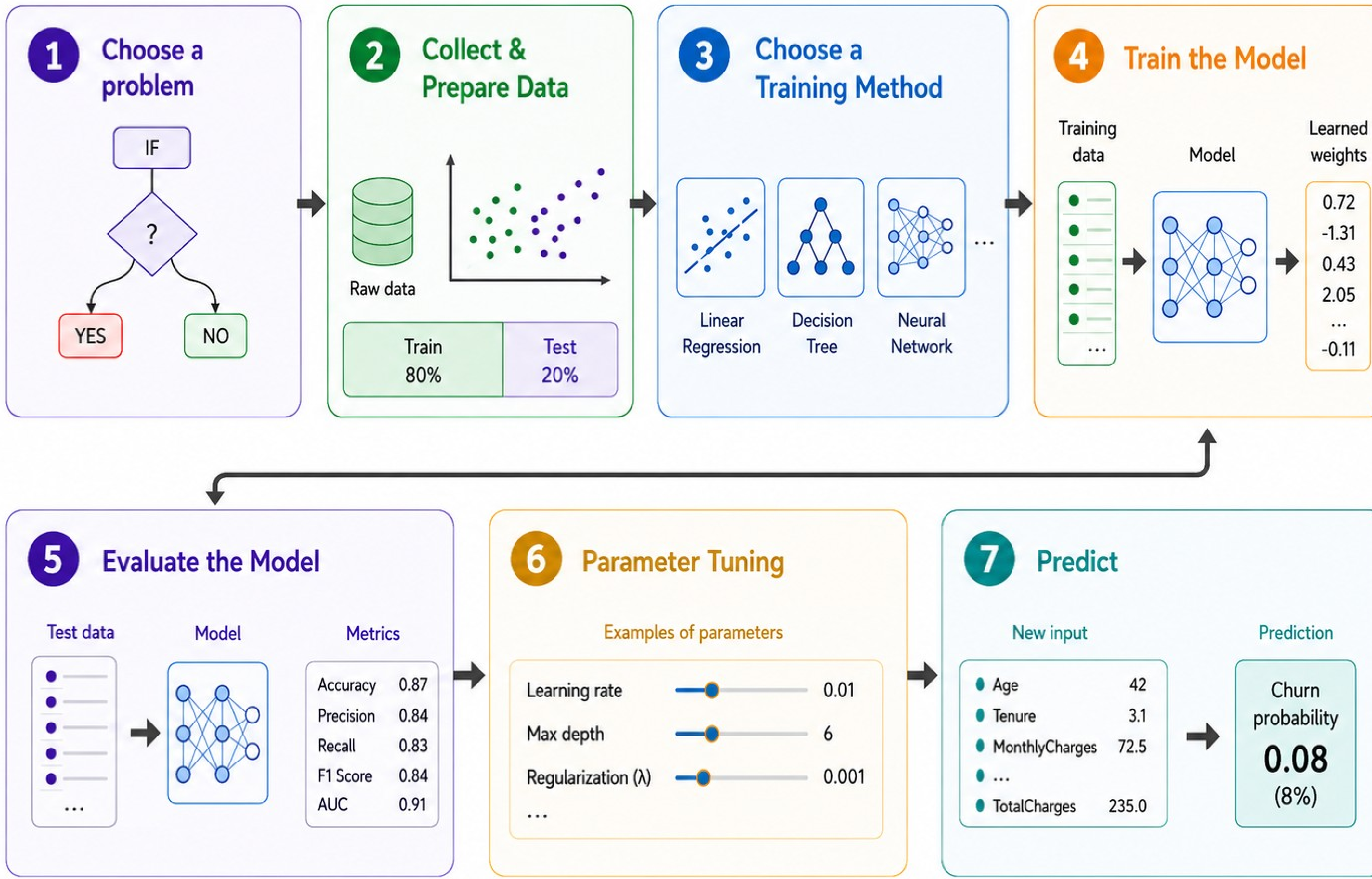



https://www.coursera.org/articles/ai-vs-deep-learning-vs-machine-learning-beginners-guide

# Four Pillars of Healthcare Audio AI (Andreas et al., 2023)



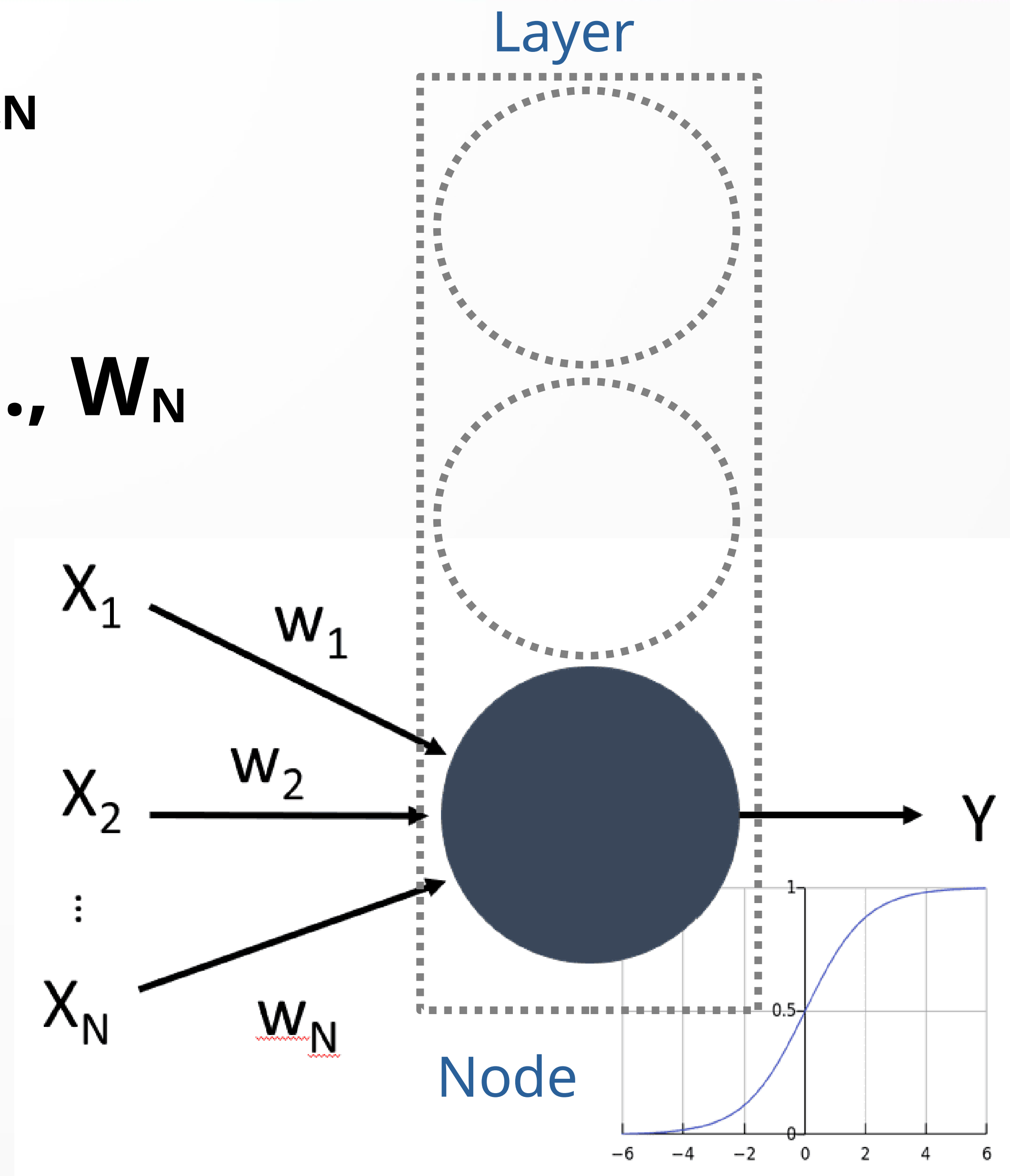
# The Craft of Creating AI Model



# Perceptron (MLP/FF)

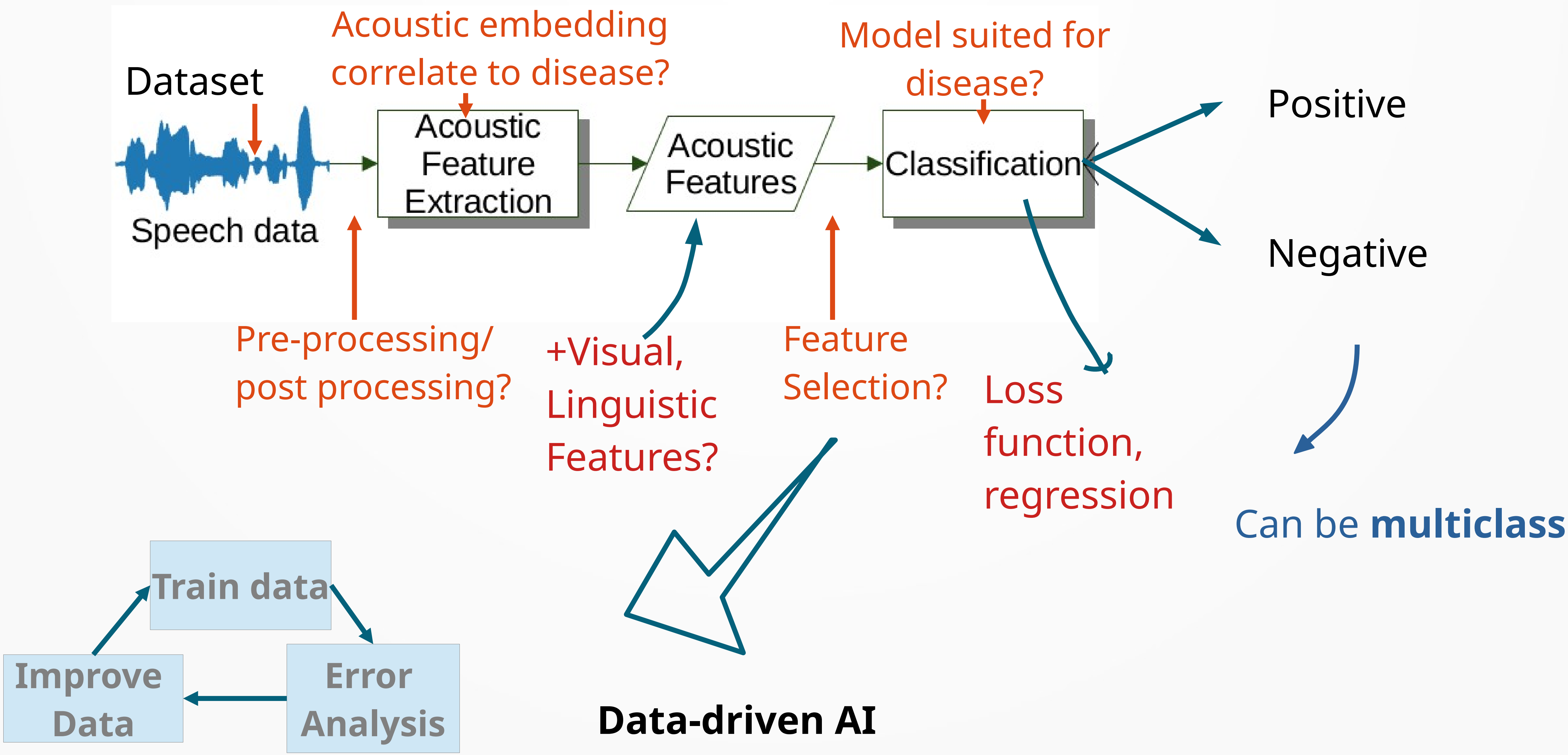
- Multiple inputs: $\mathbf{X_1, X_2, \ldots, X_N}$
- Single output: **Y**
- Series of weights: $\mathbf{W_1, W_2, \ldots, W_N}$
- Activation function: **f**

$$Y = f\left(\sum_{i=1}^{N} X_i \cdot W_i\right)$$

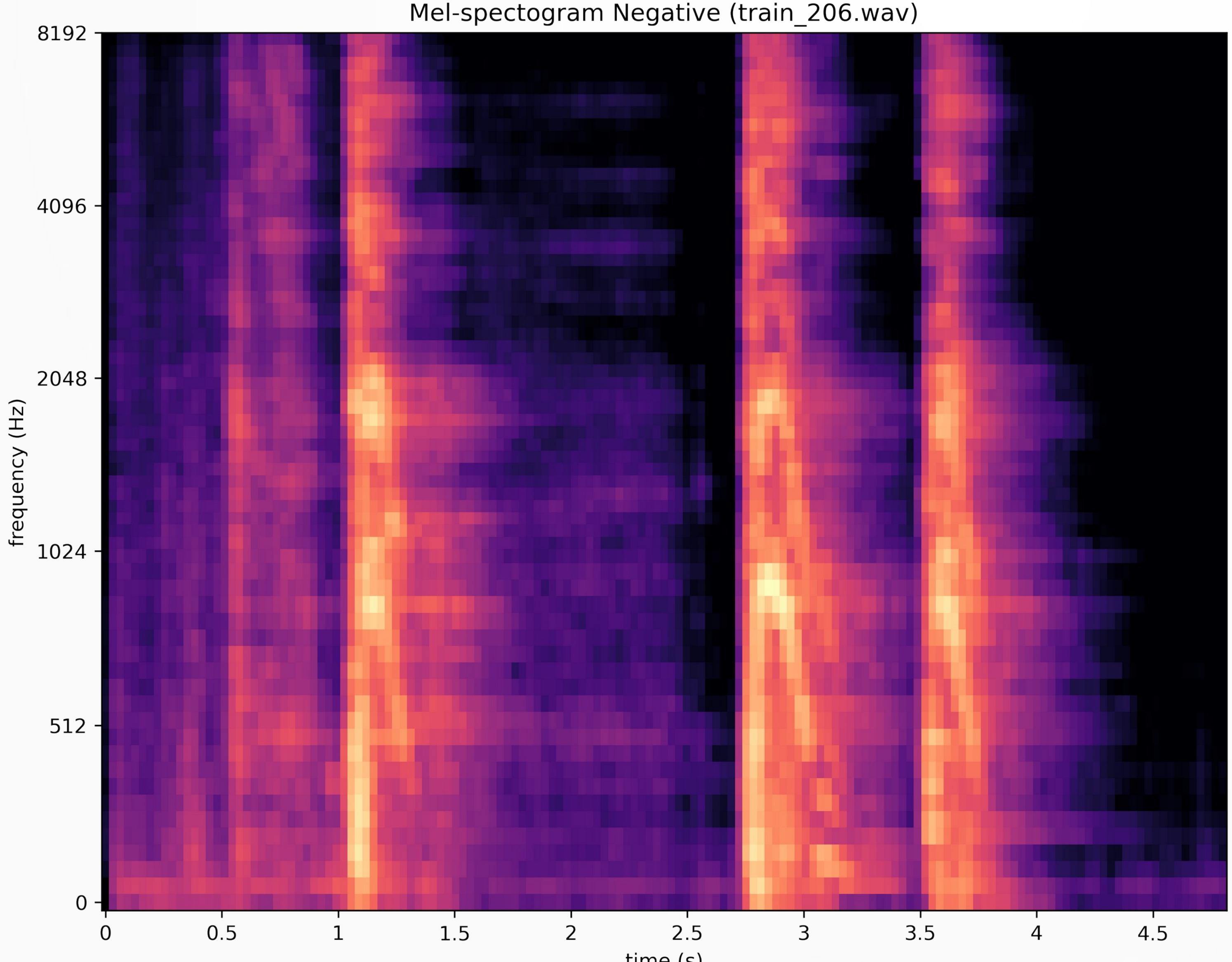


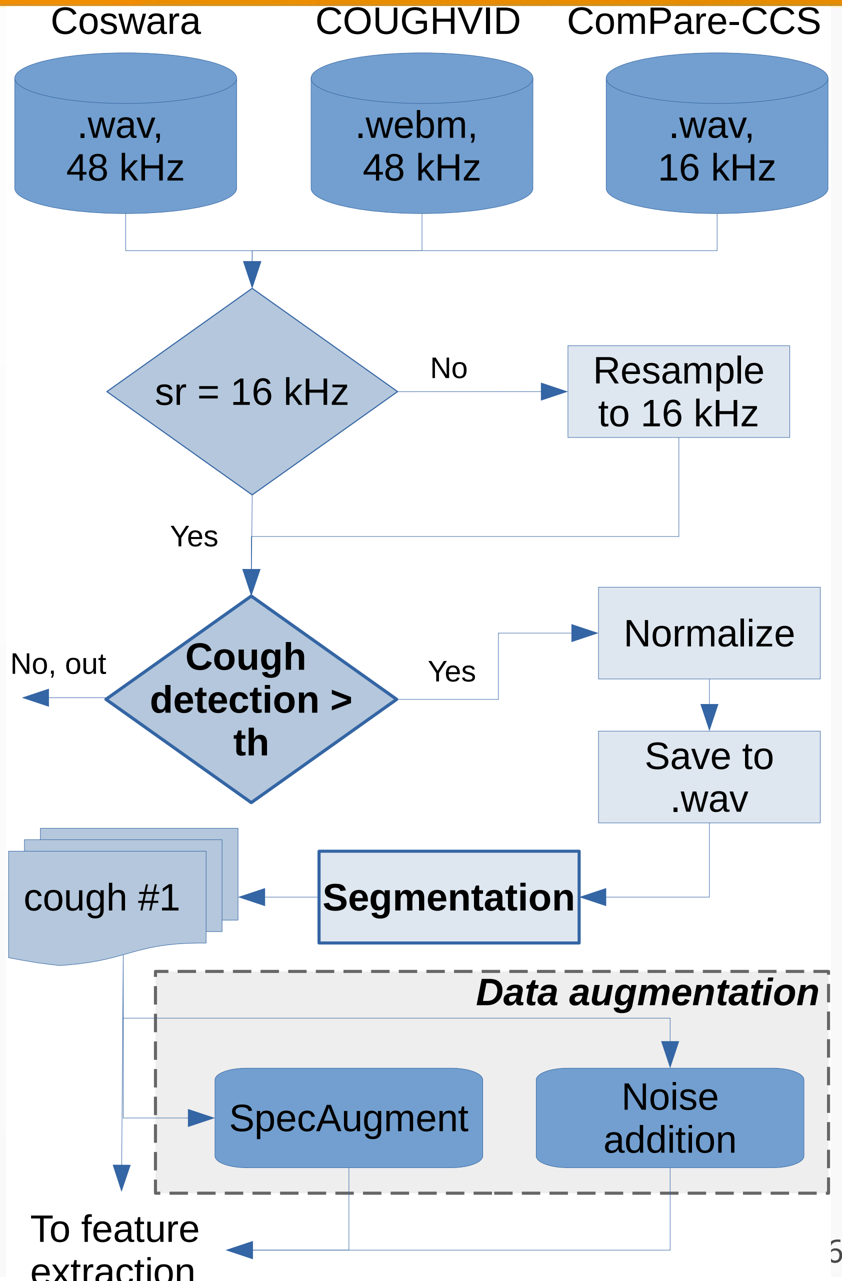
# PART I:
# Voice AI

# Typical Workflow



# Acoustic Feature Extraction




Atmaja, B. T., et.al. (2025). Cross-dataset COVID-19 transfer learning with data augmentation. International Journal of Information Technology. https://doi.org/10.1007/s41870-025-02433-z

# COVID-19 detection using CNN

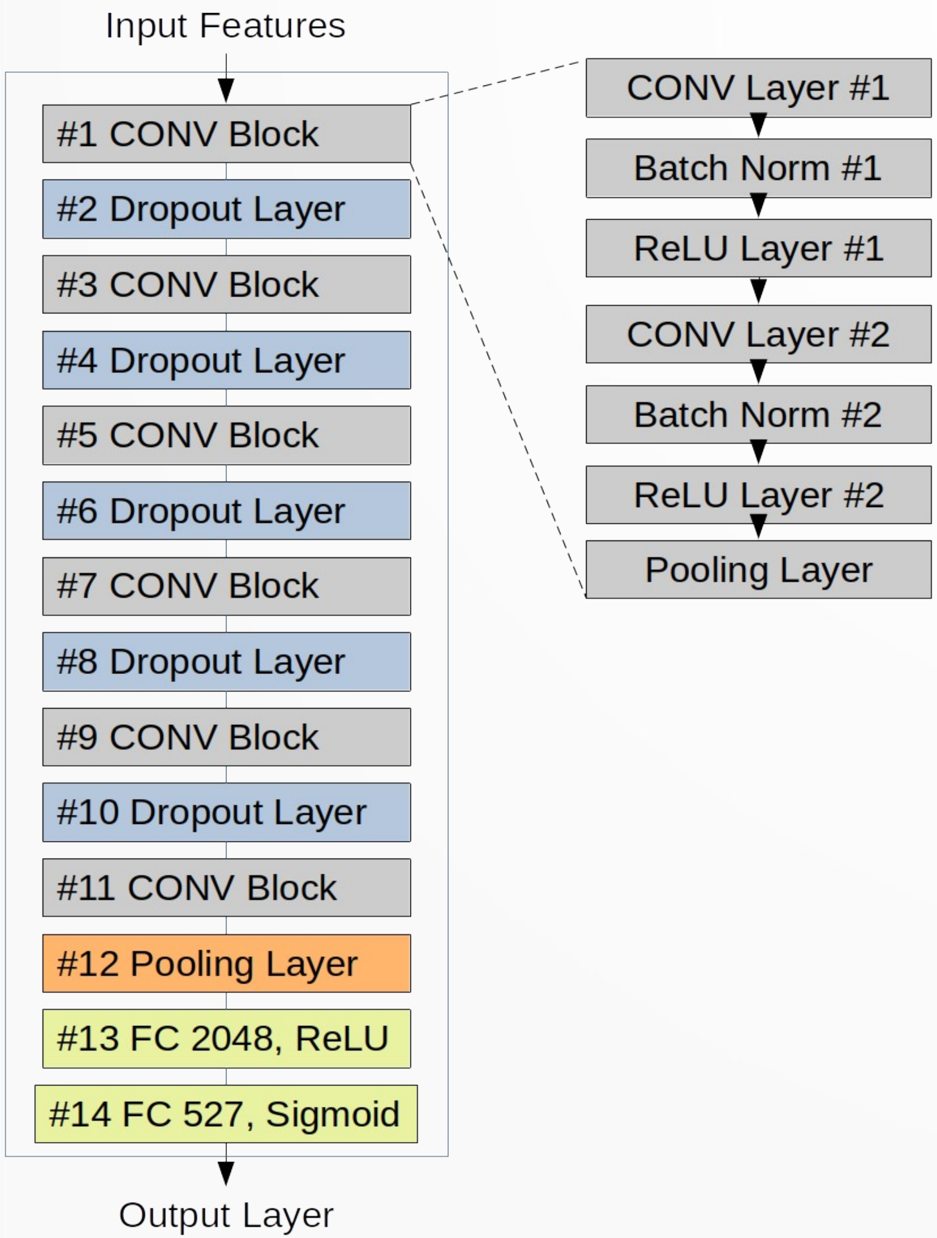



**Table 4** UA results on different data augmentation methods

| Segmentation method | Augmentation method | UA (%) |
|---|---|---|
| Hysteresis comparator | Without augmentation | 81.68 |
| | SpecAugment | **86.36** |
| | Noise addition | 81.40 |
| | SpecAugment + Noise addition | 83.19 |

| Reference | Data Aug | Feature | Classifier | Ensemble | UA |
|---|---|---|---|---|---|
| Baseline [31] | × | openSMILE, openXBOW, DeepSpctrum, AuDeep | SVM, End2You | √ | 73.90 |
| Casanova et al. [23] | √ | log mel spectrogram | CNN14 | √ | 75.90 |
| Illium et al. [49] | √ | log mel spectrogram | Vision Transformer | × | 76.90 |
| Solera-Urena et al. [50] | × | TDNN-F, VGGish, PASE+ | SVM | √ | 69.30 |
| Suyanto et al. [26] | × | log mel spectrogram | CNN14 | × | 83.19 |
| Ours | √ | log mel spectrogram | CNN14+HPO | × | **88.19** |

# Cough Segmentation

Int. j. inf. tecnol.
https://doi.org/10.1007/s41870-023-01626-8

ORIGINAL RESEARCH

## Comparing hysteresis comparator and RMS threshold methods for automatic single cough segmentations

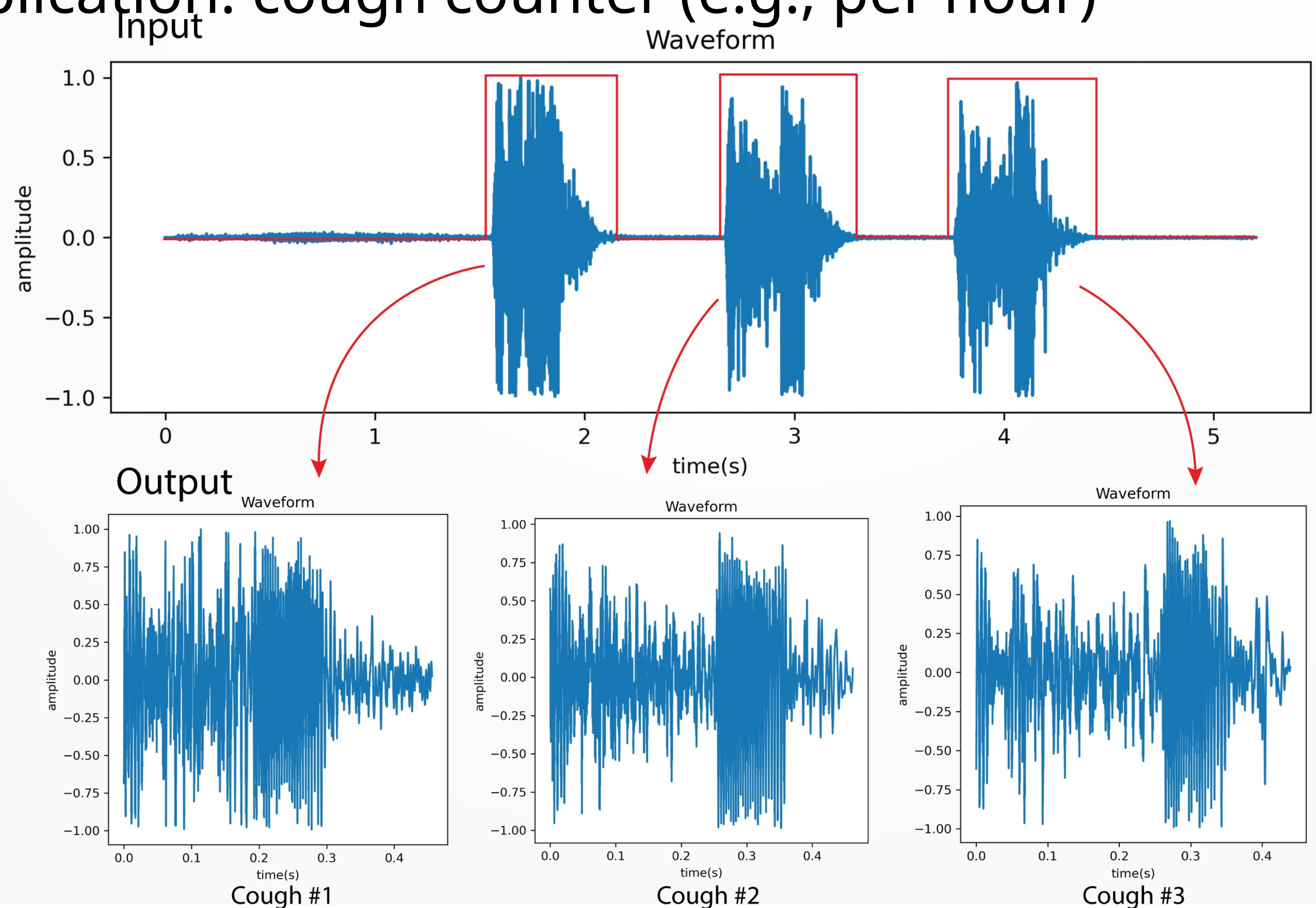
**Bagus Tris Atmaja[1,2] · Zanjabila[2] · Suyanto[2] · Akira Sasou[1]**

Received: 30 July 2023 / Accepted: 6 November 2023

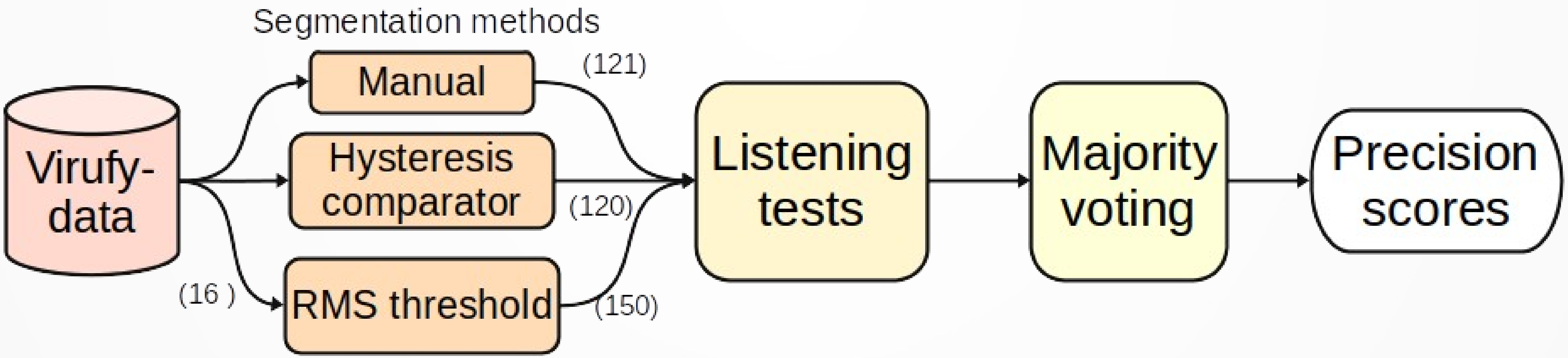
Atmaja, B. T., Zanjabila, Suyanto, & Sasou, A. (2023). Comparing hysteresis comparator and RMS threshold methods for automatic single cough segmentations. International Journal of Information Technology, 0123456789.

# Cough Segmentation

- Segmenting multiple cough into individual cough
- Application: cough counter (e.g., per hour)



# Cough Segmentation result




| Segmentation method | Fleiss' Kappa | Interpretation |
|---|---|---|
| Manual segmentation | 0.345 | fair |
| Hysteresis comparator | 0.246 | fair |
| RMS threshold | 0.486 | moderate |

| Segmentation method | N | Single-cough | Precision |
|---|---|---|---|
| Manual segmentation | 121 | 60 | 49.59% |
| Hysteresis comparator | 120 | 88 | 73.33% |
| RMS threshold | 150 | 105 | 70.00% |

# Recap: previous works on well-being

- Depression detection: Burkhardt, F., Atmaja, B. T., Derington, A., & Eyben, F. (2024). Check Your Audio Data : Nkululeko for Bias Detection. 2024 Oriental COCOSDA.
- Lauhter classification: Atmaja, B. T., Sasou, A., & Burkhardt, F. (2025). Performance-Weighted Ensemble Learning for Speech Classification. 2025 ICAIIC

| Feature | Androids | UASpeech |
|---|---|---|
| Duration | .067 | Unclear-Mixed: **1.912** |
| Age | .128 | Spastic-Athetoid: .169 |
| Gender: $\chi^2$ p | $\sim$ **0** | $\sim$ **0** |
| PESQ | .407 | Unclear-Mixed: **2.014** |
| MOS | .459 | Unclear-Athetoid: .385 |
| SDR | .019 | Unclear-Mixed: **1.169** |
| Arousal | **.666** | Spastic-Athetoid: .256 |
| Valence: C.d | .415 | Athetoid-Mixed: .289 |

**Table 1.** Results of the influence of various predicted influencing factors with respect to the target labels.

| Task-dataset, features | Mean | | Uncertainty | | UA-weighted | | WA-weighted | |
|---|---|---|---|---|---|---|---|---|
| | UA | WA | UA | WA | UA | WA | UA | WA |
| LC-Laughter os+praat | 69.4 | 65.2 | 69.4 | 65.2 | 69.4 | 65.2 | 73.0 | 69.6 |

# Pathological Speech Detection

- Voice disorder detection determines whether a voice sample contains pathology or not.
- The task is binary classification with F1-score and area under curve (AUC) were the most cited metrics
- Identification further classifies the type of voice disorder if detected as pathological, e.g., if a voice falls into one type of structural, neurogenic, functional, and psychogenic.

# Performance of different vowel

TABLE I
PERFORMANCE OF DIFFERENT VOWELS (CLASSIFIER: XGB)

| Vowels | UA | WA | F1 | F1-ma | F1-we | AUC |
|---|---|---|---|---|---|---|
| | | | os | | | |
| a | **0.771** | 0.773 | 0.817 | 0.760 | 0.777 | **0.854** |
| i | 0.609 | 0.700 | 0.799 | 0.606 | 0.662 | 0.609 |
| u | 0.754 | **0.779** | **0.831** | **0.756** | **0.778** | 0.836 |
| aiu | 0.706 | 0.742 | 0.806 | 0.710 | 0.739 | 0.819 |
| | | | hubert | | | |
| a | 0.717 | 0.753 | 0.814 | 0.722 | 0.749 | 0.717 |
| i | **0.729** | **0.763** | **0.822** | **0.734** | **0.760** | **0.729** |
| u | 0.655 | 0.695 | 0.770 | 0.658 | 0.691 | 0.656 |
| aiu | 0.575 | 0.647 | 0.751 | 0.574 | 0.626 | 0.575 |

# Performance of different features

| Features | Vowel | UA | WA | F1 | F1-ma | F1-we |
|---|---|---|---|---|---|---|
| os | /a/ | 0.771 | 0.773 | 0.817 | 0.760 | 0.777 |
| praat | /a/ | **0.806** | **0.784** | 0.815 | **0.778** | **0.789** |
| w2v | /i/ | 0.732 | 0.758 | 0.815 | 0.733 | 0.757 |
| hubert | /i/ | 0.729 | 0.763 | **0.822** | 0.734 | 0.760 |
| wavlm | /u/ | 0.714 | 0.753 | 0.816 | 0.720 | 0.748 |
| ft-w2v | /i/ | 0.721 | 0.758 | 0.819 | 0.727 | 0.754 |
| ft-hubert | /i/ | 0.682 | 0.663 | 0.704 | 0.657 | 0.671 |
| ft-wavlm | /i/ | 0.680 | 0.726 | 0.798 | 0.686 | 0.719 |

# Performance of ensemble learning

| Method | UA | WA | F1 | F1-ma | F1-we | AUC |
|---|---|---|---|---|---|---|
| early fusion | | | | | | |
| os+praat | 0.780 | 0.789 | 0.833 | 0.774 | 0.791 | 0.872 |
| w2v+h+w | 0.731 | 0.753 | 0.808 | 0.730 | 0.753 | 0.809 |
| late fusion | | | | | | |
| os+praat | 0.797 | 0.795 | 0.833 | 0.784 | 0.798 | **0.876** |
| w2v /a/+/u/ | 0.752 | 0.789 | 0.844 | 0.761 | 0.785 | 0.831 |
| os+praat+w2v | 0.814 | 0.816 | 0.852 | 0.804 | 0.818 | 0.869 |
| os+praat+h | **0.841** | **0.842** | **0.874** | **0.831** | **0.842** | 0.867 |

# Pathological Speech Benchmark

## F1-score

| Method | F1 score |
|---|---|
| CAR-HMM [16] | 0.7527 |
| modidifed CPC [16] | 0.7421 |
| os /u/ (ours) | 0.8306 |
| hubert /i/ (ours) | 0.8221 |
| os+praat+hubert (ours) | **0.8739** |

## AUC

| Method | Vowel/Phrases | Augmentation | AUC |
|---|---|---|---|
| Perturbation [1] | /a/ | No | 0.78 |
| MFCC raw speech [1] | phrases | No | 0.86 |
| MFCC extracted vowel [1] | phrases | No | 0.84 |
| ComParE [3] | phrases | No | 0.72 |
| wav2vec2 + CNN [14] | phrases | No | 0.87 |
| Transformers [30] | phrases + /aiu/ | Yes | 0.91 |
| os+praat + XGB (ours) | /a/ | No | 0.88 |

# Dementia Prediction From Speech Signal Using Optimized Prosodic Features

Bagus Tris Atmaja, Sakriani Sakti
Presented at APSIPA-ASC 2025, Singapore

## Why we need to detect dementia as early as possible:

- Early detection of dementia, cognitive decline, is vital for enhancing patient care and management
- Conventional clinical assessments and cognitive tests require expert clinicians and are time-consuming
- Developing rapid and reliable in-house detection methods is highly desirable
- We proposed optimized prosodic features (15 features), with a focus on pause-related features, for dementia classification

# Proposed Acoustic Features

- pause_lognorm_mu: Location parameter (μ) of the log-normal distribution fitted to pause duration[1]
- pause_lognorm_sigma: Shape parameter (σ) of the log-normal distribution fitted to pause duration[1]
- pause_lognorm_ks_pvalue: P-value from Kolmogorov-Smirnov test for goodness of fit of the log-normal distribution[1]
- pause_mean_duration: Mean duration of pauses between speech segments [2]
- pause_std_duration: Standard deviation of pause duration[2]
- pause_cv: Coefficient of variation (CV) of pause duration (std/mean)
- Proportion of pause duration: Proportion of total pause duration relative to speaking time[2]

[1] P. Pastoriza-Domınguez, I. G. Torre, F. Di´eguez-Vide, et al., Speech pause distribution as an early marker for Alzheimer's disease, Jan. 2021

[2] R. Haulcy and J. Glass, "CLAC: A speech corpus of healthy english speakers," in Proc. Annu. Conf. Int. Speech Commun. Assoc. INTERSPEECH, vol. 1, 2021, pp. 201–205,

# Dementia Datasets

- **DementiaNet**[1] contains public figures samples (WAV) with a confirmed dementia diagnosis and other public figures with no cognitive decline (control) over the age of eighty.
- **DementiaBank**[2] is a shared database of multi-media interactions for the study of communication in dementia. We analyzed subset of DementiaBank namely **Pitt Corpus** in MP3 format.
- **Mixed** of both datasets above

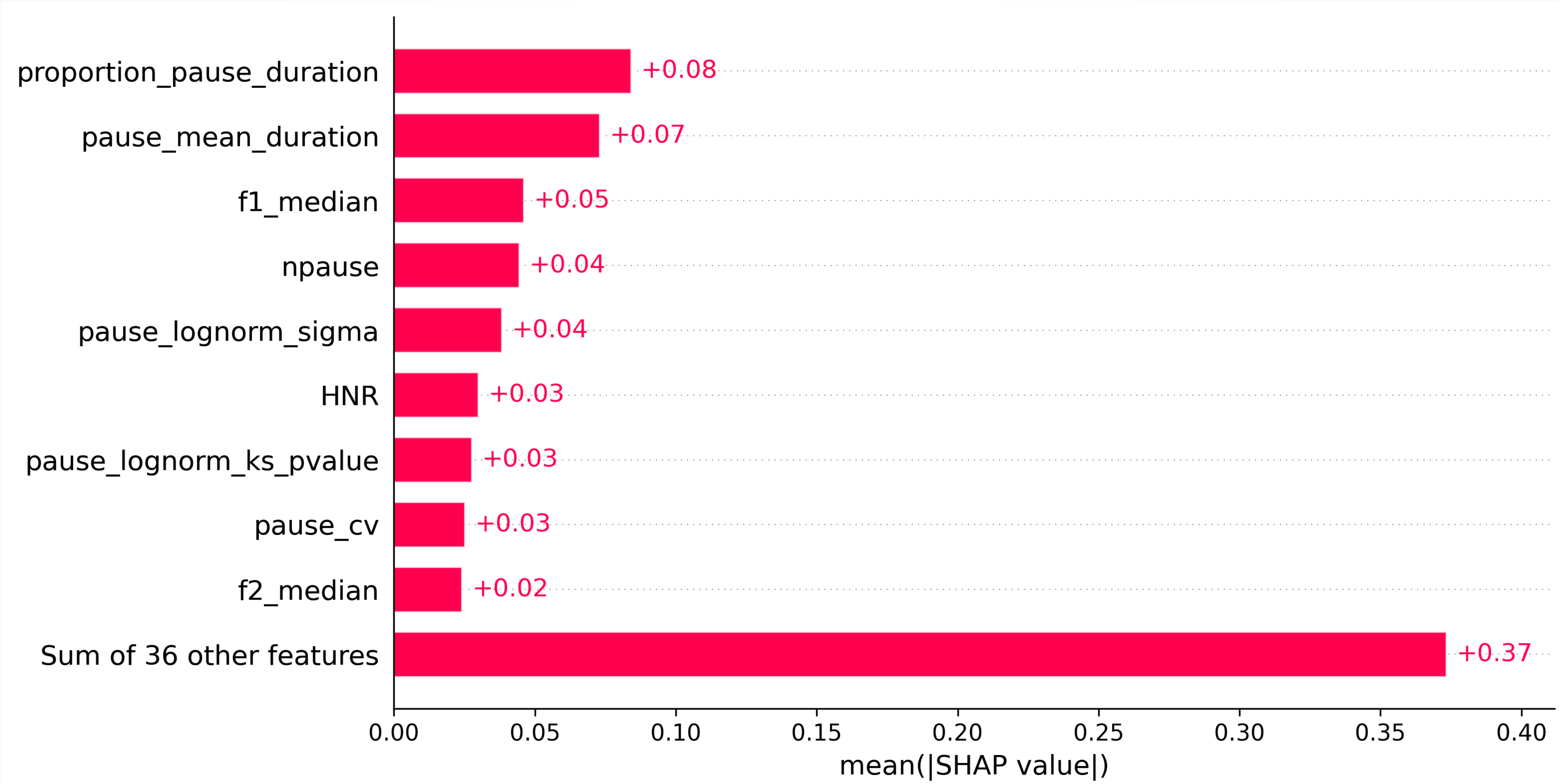
Model → XGB        Feature importance → SHAP, XGB

[1] https://github.com/shreyasgite/dementianet

[2] https://talkbank.org/dementia/

[2] https://www.tensorflow.org/datasets/catalog/dementiabank

# Results: DementiaNet



# Results: DementiaBank & Mixed

DementiaBank

| Feature | XGB Importance |
|---|---|
| f3_median | 0.07530 |
| pause_lognorm_ks_pvalue | 0.00803 |
| pause_lognorm_mu | 0.00550 |
| npause | 0.00537 |
| duration | 0.00417 |
| nsyll | 0.00290 |
| f1_mean | 0.00278 |
| localabsoluteJitter | 0.00248 |
| proportion_pause_duration | 0.00236 |
| ppq5Jitter | 0.00115 |
| meanF0Hz | 0.00066 |
| ASD_speakingtime_nsyll | 0.00042 |
| pF | 0.00042 |
| apq3Shimmer | 0.00030 |
| pause_mean_duration | 0.00012 |

Mixed Dataset

| Feature | XGB Importance |
|---|---|
| f1_median | 0.02894 |
| proportion_pause_duration | 0.01306 |
| f3_median | 0.00915 |
| pause_lognorm_ks_pvalue | 0.00867 |
| meanF0Hz | 0.00697 |
| ASD_speakingtime_nsyll | 0.00605 |
| f1_mean | 0.00540 |
| stdevF0Hz | 0.00516 |
| npause | 0.00387 |
| pF | 0.00359 |
| localdbShimmer | 0.00343 |
| f3_mean | 0.00294 |
| pause_lognorm_mu | 0.00242 |
| f4_mean | 0.00198 |
| HNR | 0.00161 |

# Results: 45 vs. 15 features

| Dataset | UA | WA | F1-score | n_feat-model |
|---|---|---|---|---|
| | | Baseline | | |
| DementiaNet | 0.746 | 0.771 | 0.820 | 45 |
| DementiaBank | 0.669 | 0.666 | 0.707 | 45 |
| Mixed dataset | 0.611 | 0.601 | 0.619 | 45 |
| | | Optimized | | |
| DementiaNet | 0.796 | 0.812 | 0.847 | 15-shap |
| DementiaBank | 0.748 | 0.745 | 0.776 | 15-xgb |
| Mixed dataset | 0.721 | 0.720 | 0.712 | 15-xgb |

**Benchmark:**

| Method | n_feat | Dataset | UA | WA | F1-score |
|---|---|---|---|---|---|
| PRAAT [18] | 7 | Elderly | - | 0.582 | 0.621 |
| IS10 [19] | 75 | Pitt Corpus | - | 0.731 | - |
| MFCC++ [20] | 44 | Pitt Corpus | - | 0.876 | 0.875 |
| eGeMAPS [21] | 88 | ADReSSo | 0.730 | 0.746 | 0.750 |
| This study | 15 | Pitt Corpus | 0.748 | 0.745 | 0.776 |
| This study | 15 | DementiaNet | 0.796 | 0.812 | 0.847 |

# Library Ecosystem + Demo

- AudioKit: Basic toolkit to support Audio Work
- **Coughkit: Detect, Segment, and Count Cough**
- **Nkululeko: Machine learning speaker characteristics, support bimodal (audio+text)**
- **MultiBench: Mutmodal AI toolkit**

```
[FEATS]
# Combine BERT linguistic features with OpenSMILE acoustic features
type = ['bert', 'os']
os.set = eGeMAPSv02
```

# PART I: Multimodal AI

**Single modalities are often noisy, ambiguous, or incomplete — combining them produces more robust and accurate systems.**

Mostly taken from: https://bagustris.github.io/multisensory/computation.html

# Three Foundational Principles

Every multimodal problem is shaped by three properties of the data:

- **Modality Heterogeneity**: How different are the modalities from each other?
- **Modality Connections**: How are elements across modalities related?
- **Modality Interactions**: How do modalities jointly influence the task?

# Principle 1 – Modality Heterogeneity

| Dimension | Description | Example |
|---|---|---|
| **Element** | Basic unit of the modality | CT voxel vs. clinical-note word vs. laboratory value |
| **Distribution** | Frequency of elements | Common diagnoses greatly outnumber rare diseases |
| **Structure** | How elements compose | MRI is spatial; ECG is sequential; EHR data is tabular |
| **Information** | Total content per modality | A CT scan is dense; a diagnosis code is sparse |
| **Noise** | Type of corruption | Motion artifacts, ECG interference, or missing lab results |
| **Task relevance** | How useful the modality is | For heart-attack detection, ECG and troponin are more relevant than a clinical photograph |

***These modalities describe the same patient but differ in representation, scale, noise, and diagnostic value.***

# Principle 2 – Modality Connections

- How are elements in different modalities related?

- Statistical association: co-occurrence (lip movement ↔️ speech sound)
- Statistical dependence: causal or confounding relationship (emotion → voice pitch AND facial expression)
- Semantic correspondence: same concept, different surface form ("dog" ↔️ image of dog)
- Semantic relations: broader, relational meaning ("red" ↔️ visual redness, danger, emotion)

# Principle 3 – Modality Interactions

$$I(M_1; M_2; Y) = \underbrace{R}_{\text{redundancy}} + \underbrace{U_1 + U_2}_{\text{uniqueness}} + \underbrace{S}_{\text{synergy}}$$

Example: Task: predict sentiment from a video clip

**"This is great"  (words = positive)**

**[flat tone]  (audio = neutral)**

**[frown]  (face = negative)**

- If words dominate → false positive
- Integrating all three → recognizes **sarcasm**
- This is a case of unique information in the non-verbal channels and synergistic interaction across all three modalities.

# Six Core Technical Challenges

| Challenge | Core Questions |
|---|---|
| Representation | How do we encode and combine multimodal data? |
| Alignment | How do we link elements across modalities? |
| Reasoning | How do we compose multimodal knowledge? |
| Generation | How do we produce coherent multimodal output? |
| Transference | How do we transfer knowledge across modalities? |
| Quantification | How do we measure heterogeneity and interactions? |

Representation sub-challenges:

Fusion — integrate two or more modalities into a joint representation

$$z_{mm} = f(x_1, x_2, \ldots, x_M)$$

Coordination — keep modalities separate but align them in a shared space

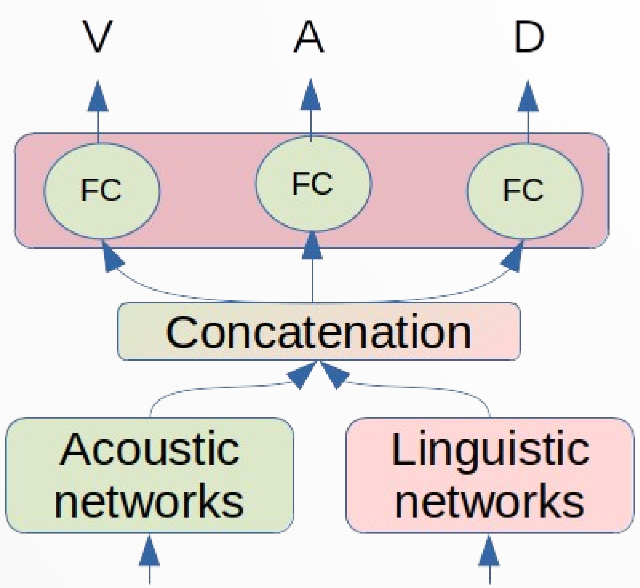
$$\mathrm{sim}(\phi_1(x_1), \phi_2(x_2)) \uparrow \text{ for matched pairs}$$

Fission — decompose into disjoint factors (modality-specific + shared)

$$z = [z_{\text{shared}},\; z_{\text{modal-1}},\; z_{\text{modal-2}}]$$

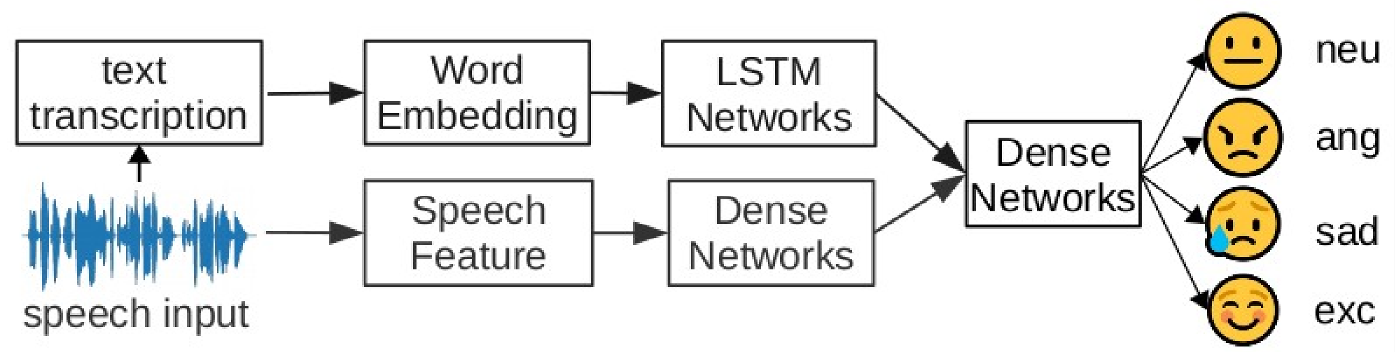
# Example of past works

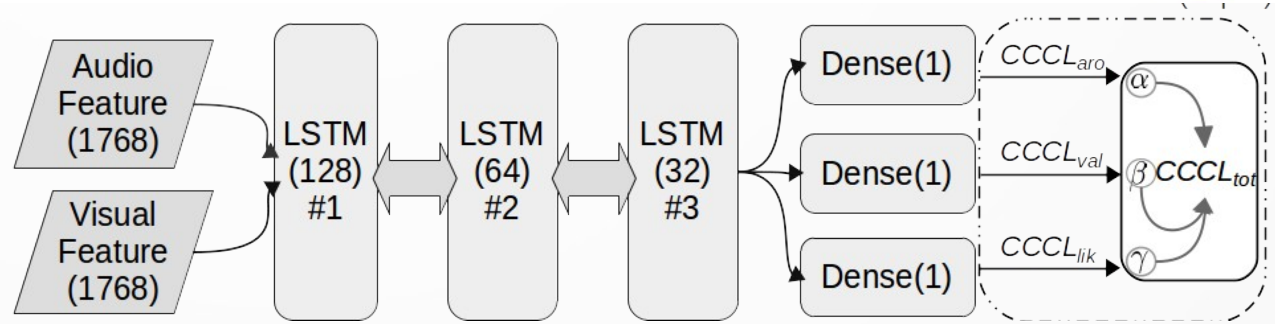
*Unimodalities are often noisy, ambiguous, or incomplete; multimodal produces more robust and accurate systems.*




Acoustic <u>features:</u>
pAA LLD
pAA HSF
GeMAPS LLD
GeMAPS HSF

Linguistic <u>features:</u>
WE
word2vec
GloVe
FastText
BERT




Papers:

1. Atmaja, B. T., & Akagi, M. Multitask learning and multistage fusion for dimensional audiovisual emotion recognition. In ICASSP 2020.
2. Atmaja, B. T., & Akagi, M. (2021). Two-stage dimensional emotion recognition by fusing predictions of acoustic and text networks using SVM. Speech Communication.

# Hands-On

- Linear Regression with Diabetes Dataset
- Breast Cancer Classification
- CoughKit
- MultiBench Toolkit

# Takeaway messages

- **AI is superior to human for reasoning and finding pattern** and prune to small error
- **Some models** work on **some cases**, others didn't
- **Robustness and generalization** is main goal instead of performances only
- **Large data** matters (Effective!!)
- Next: Demo